

MIDLAND BARSTOW, MI, USA

WMO: 720629

Lat: 43.663N

Lon: 84.261W

Elev: 194

StdP: 99.01

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00216

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.5	-16.1	-23.4	0.5	-18.0	-20.2	0.6	-15.2	9.3	-2.7	8.5	-3.8	0.8	220	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	32.1	22.8	30.2	21.6	28.8	20.9	24.1	29.5	23.1	28.0	22.3	26.8	3.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.7	26.7	21.6	16.7	25.7	20.9	15.9	25.0	73.5	29.6	69.5	27.9	66.2	26.7	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.7	7.7	6.9	-24.6	34.4	3.9	1.8	-27.3	35.7	-29.6	36.7	-31.7	37.7	-34.5	39.1
			-24.8	26.0	3.8	1.2	-27.5	26.9	-29.8	27.5	-31.9	28.2	-34.7	29.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-5.1	-4.4	0.8	7.2	14.5	18.8	21.5	20.4	16.6	10.2	3.6	-1.3
	DBStd	10.56	6.12	6.23	6.15	4.90	4.92	3.75	3.34	3.02	4.43	4.69	4.89	4.99
	HDD10.0	1901	469	404	294	111	12	0	0	0	2	57	200	352
	HDD18.3	3842	727	638	544	336	142	39	9	14	86	254	442	609
	CDD10.0	1397	0	0	8	26	151	263	356	323	199	64	8	1
	CDD18.3	296	0	0	0	1	22	51	107	79	33	3	0	0
	CDH23.3	3095	0	0	5	23	281	573	1156	730	293	33	0	0
	CDH26.7	911	0	0	0	3	71	168	402	188	76	3	0	0
Wind	WSAvg	2.6	3.0	3.1	3.0	3.4	2.8	2.3	1.9	1.8	1.9	2.6	2.8	2.8
Precipitation	PrecAvg	728	39	32	52	71	73	75	62	83	85	61	63	45
	PrecMax	1004	106	99	144	138	186	232	120	324	466	139	169	131
	PrecMin	476	6	2	9	22	10	18	8	22	0	16	13	3
	PrecStd	115	24	23	30	31	39	41	30	49	72	28	36	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	14.3	24.9	27.1	31.2	32.8	34.3	32.4	32.5	27.0	20.5	14.2
		MCWB	10.5	10.9	17.0	16.1	20.8	22.6	24.0	22.4	22.8	18.3	14.1	12.1
	2%	DB	7.7	9.1	17.8	22.3	28.8	30.6	32.4	30.2	28.8	23.6	16.4	10.1
		MCWB	5.5	6.6	12.4	14.3	19.7	20.9	23.5	22.0	21.7	17.4	11.9	8.0
	5%	DB	4.4	5.8	13.7	18.9	26.8	28.6	30.6	28.8	26.7	20.9	13.8	7.3
		MCWB	3.0	3.4	9.8	12.1	17.8	20.6	22.2	21.2	20.0	16.3	9.9	5.8
	10%	DB	2.7	3.7	10.1	16.3	24.3	26.8	28.8	27.4	24.8	18.5	11.2	5.0
		MCWB	1.5	1.6	6.4	10.3	16.9	19.3	21.1	20.3	19.0	15.1	8.6	3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	11.4	17.4	17.8	22.5	24.4	25.9	24.4	24.1	20.8	15.2	12.7
		MCDB	11.1	13.0	25.5	23.7	29.0	30.4	32.2	28.8	29.5	23.6	18.1	13.7
	2%	WB	5.6	6.7	13.8	15.0	20.8	22.7	24.7	23.2	22.8	19.1	12.4	8.4
		MCDB	7.1	9.2	16.8	20.5	25.8	28.3	30.5	27.7	27.4	21.6	15.8	10.0
	5%	WB	3.3	3.5	10.3	13.0	19.7	21.6	23.5	22.5	21.5	17.1	10.4	5.9
		MCDB	4.5	5.4	13.3	17.9	24.5	26.8	28.6	26.9	24.8	20.1	13.2	7.2
	10%	WB	1.7	1.7	6.4	10.8	17.9	20.4	22.5	21.5	20.3	15.1	8.6	3.4
		MCDB	2.7	3.4	10.0	15.7	22.8	24.8	26.8	25.8	23.5	17.7	11.4	4.8
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.9	10.1	11.4	12.8	12.5	12.8	12.3	12.4	10.8	8.8	6.4
		MCDBR	8.4	10.1	15.1	16.8	15.6	14.3	14.2	13.4	13.6	14.5	12.7	9.2
		MCWBR	7.4	7.9	10.1	9.5	7.7	6.6	6.4	6.1	6.9	8.4	8.8	7.8
	5% WB	MCDBR	8.3	9.8	13.3	13.6	13.1	11.9	12.0	11.4	11.3	11.8	11.6	8.6
		MCWBR	7.6	8.3	9.7	9.3	7.1	6.4	6.1	5.8	6.4	8.4	9.1	7.8
Clear-Sky Solar Irradiance	taub	0.289	0.318	0.328	0.363	0.388	0.417	0.406	0.398	0.375	0.339	0.311	0.282	
	taud	2.374	2.286	2.396	2.327	2.310	2.256	2.295	2.312	2.375	2.473	2.493	2.447	
	Ebn at Noon	840	869	913	901	884	856	861	856	849	841	813	819	
	Edh at Noon	78	101	104	121	126	134	128	121	105	84	70	66	
All-Sky Solar Radiation	RadAvg	1.39	2.28	3.60	4.56	5.37	6.02	6.08	5.17	4.13	2.57	1.55	1.10	
	RadStd	0.13	0.27	0.39	0.39	0.49	0.35	0.30	0.24	0.39	0.27	0.20	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page